Mohammad Umar Haris

Data Scientist

1590, Lal Kuan Bazaar, Delhi-110006 | +91 8447582440

mohdomarharis@gmail.com | linkedin.com/in/mohdomarharis | www.umarharis.com

Professional Summary

Data Scientist with 5+ years of experience in analytics, machine learning, and Generative AI application development across insurance, logistics, and pharmaceutical domains. Hands-on in analysing large datasets, writing and optimising SQL, and building, deploying, and monitoring ML models using Python and Azure (Databricks, Azure SQL, Azure ML, Azure OpenAI), MLflow, and FastAPI. MSc Data Science (Distinction) with applied expertise in predictive modelling, NLP, forecasting, and CI/CD automation via GitHub Actions. Immediate joiner.

Core Skills

- **Generative AI & LLMs:** LLM API Integration, Prompt Engineering, Azure OpenAI, Text Summarisation Pipelines, Copilot Integration
- **ML & Statistics:** Predictive Modelling, Classification, Regression, Time Series Forecasting, NLP, Feature Engineering, Deep Learning (PyTorch, TensorFlow), A/B Testing, Hypothesis Testing, Model Evaluation
- **MLOps & Deployment:** MLflow, FastAPI, GitHub Actions (CI/CD), Azure ML, Docker, Model Versioning, Model Monitoring, REST API Deployment
- **Programming & Analysis:** Python (Pandas, NumPy, Scikit-learn), SQL (Azure SQL, PostgreSQL, MySQL), R
- **Cloud & Data Engineering:** Azure Databricks, Azure Data Factory, Azure SQL, ETL Pipelines
- **BI & Visualisation:** Power BI, Tableau, Google Data Studio
- **Practices:** Agile (Scrum), Stakeholder Management, Cross-functional Collaboration, Model Validation

Professional Experience

Data Scientist **Sept 2023 – Jan 2026**

System Engineering Ltd, London, UK (Oct 2022 – Jan 2026)

- Built and deployed a production GenAI pipeline on Azure for a UK insurance provider, using Azure Speech-to-Text with speaker diarisation and Azure OpenAI to summarise calls and extract intent, sentiment, and key case details.
- Developed a GenAI orchestration harness with prompt engineering, output validation, and retry loops for malformed LLM responses, generating recovery options and recommended actions that replaced manual case review.
- Served models and LLM pipeline outputs as REST APIs with FastAPI, managed experiment tracking and model versioning in MLflow, and automated testing and deployment through GitHub Actions CI/CD on Azure.
- Built Azure Databricks (Medallion architecture) and Azure SQL ETL pipelines over operational and CRM data, reducing reporting turnaround by 35%.
- Analysed large CRM and operational datasets using Python and SQL to surface patterns, trends, and actionable insights, and engineered features and statistical models to improve case response efficiency.
- Developed predictive forecasting models for demand and recovery operations, improving planning accuracy by 22%, and monitored model performance to refine them as data evolved.
- Partnered with operations and business stakeholders to translate requirements into data-driven solutions, integrating a Microsoft Copilot chatbot into recovery workflows and delivering Power BI KPI dashboards.
- Automated data validation and reporting, saving 10+ analyst hours every week.

Junior Data Scientist (Part-time) **Oct 2022 – Aug 2023**

System Engineering Ltd, London, UK (promoted to Data Scientist, Sept 2023)

- Contributed to a classification-based product recommendation engine, helping frame recommendations as a binary purchase-probability problem to support more personalised targeting.
- Built and evaluated classical ML models in scikit-learn (Logistic Regression, Random Forest, XGBoost), supporting a baseline that improved on the legacy rule-based approach.
- Supported customer and product segmentation using K-Means and DBSCAN clustering, feeding the recommendation model.
- Engineered behavioural and transactional features from SQL data using Pandas and NumPy, handling missing values, skewed distributions, and categorical encoding (one-hot and target encoding).

- Supported model evaluation under class imbalance using SMOTE and class weighting, tracking ROC-AUC and F1-score, and assisted the product team with A/B test analysis of click-through rate.

Associate Analyst **Dec 2021 – Sept 2022**
Cognizant Technology Solutions, India

- Engineered data quality and validation pipelines for Google Ads content policy, processing 7,200+ monthly records at 99% accuracy and systematically resolving 100+ weekly extraction anomalies — achieving top 5 data accuracy performance across 30+ teams.
- Automated policy validation workflows for a financial services client using rule-based logic and structured data checks, eliminating manual review overhead and implementing process controls that reduced data discrepancy rates by 30%.
- Designed and deployed KPI dashboards in Google Data Studio and Google Sheets with automated data feeds, enabling real-time governance reporting and performance tracking across multiple business units.
- Executed A/B testing frameworks on operational workflows driving 15–20% conversion uplift; developed an Excel-based decision modelling tool with structured logic that reduced policy evaluation time by 30%, standardising a previously inconsistent manual process.

Data Analyst (Part-time) **Jan 2020 – Dec 2021**
Inletware Ltd

- Performed end-to-end analysis of pharmaceutical e-commerce data using SQL, cleaning and transforming raw transactional and inventory datasets to support demand planning and operational reporting.
- Built time-series forecasting models to anticipate product demand and reduce stockouts, helping the team plan inventory more accurately across key product categories.
- Designed and maintained 30+ Power BI and Tableau dashboards tracking sales, inventory, and fulfilment metrics, improving reporting efficiency by ∼20% and giving stakeholders self-serve visibility into KPIs.
- Consolidated data from Azure SQL and internal sources into clean, analysis-ready tables, and automated recurring reports to replace manual spreadsheet preparation.

Selected ML / Data Science Projects

- **Insurance Fraud Detection** *(Python, XGBoost, Scikit-learn, PCA, SMOTE)* — Benchmarked multiple classifiers with XGBoost as the best performer; applied PCA, handled class imbalance with SMOTE, and tuned the decision threshold beyond the default to optimise precision.
- **Anticancer Molecule Activity Prediction (MSc Dissertation)** *(Python, Scikit-learn, RDKit)* — Classified anticancer activity across 26,391 molecules using PaDEL fingerprints and Lipinski descriptors; compared Logistic Regression, Random Forest, and neural networks with class weights, SMOTE, and upsampling.
- **Demand Forecasting & Inventory Optimisation** *(Python, time-series models, SQL, Power BI)* — Forecasting models on pharmaceutical e-commerce data to prevent stockouts and improve planning accuracy.
- **Gout Disease NLP Classification** *(Python, TF-IDF, Scikit-learn)* — NLP pipeline classifying gout-related clinical text using TF-IDF vectorisation, supervised models, and hyperparameter tuning.

Education

MSc Data Science and its Applications — Distinction 2022 – 2023
University of Essex, UK

Bachelor of Pharmacy (B.Pharm) — First Class 2017 – 2021
Jamia Hamdard University, India